

RESEARCH ARTICLE OPEN ACCESS

Data-Driven Design of Epoxy Composites: Heat Resistance Prediction Using Machine Learning Algorithms

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Received: 17 December 2025 | **Revised:** 11 March 2026 | **Accepted:** 9 April 2026

Academic Editor: Mohammad Rezwan Habib

Keywords: composite | fillers | heat resistance | machine learning algorithms | mechanical properties | polymer | prediction | transport

ABSTRACT

The paper presents the results of studies of the mechanical properties of modified and dispersion-filled epoxy composite materials: destructive stresses and modulus of elasticity in bending, impact toughness, adhesive strength, residual stresses and heat resistance. At the initial stage, we applied the analysis of a set of properties only for training ensemble machine learning (ML) algorithms. Based on experimental data, we then considered the problem of predicting the values of the variable heat resistance depending on the values of the mechanical properties of composites. To solve the prediction problem, we used various ML algorithms: AdaBoost, bagging, decision trees, gradient boost, random forest, stacking, voting, ridge regression, support vector machine (SVM) and K-nearest neighbour (k -NN), which establish the dependence of the values of the variable heat resistance on the values of the above variables. To ensure the reliability of the results and minimise dependence on the specifics of individual algorithms, a comprehensive approach was used, combining ML and mathematical statistics methods. The results of solving two important tasks have been obtained: the first task is to predict the values of the heat resistance variable based on the values of the mechanical properties of materials, and the second task is to determine the importance of the influence of the values of all variables on the value of the predicted heat resistance variable. Based on the results of the first task, it can be stated that all models (ML algorithms) obtained in the work for prediction are highly effective since their MAPE has low values on the test sample and, therefore, can be used by experimenters to reduce the costs of various types of resources for conducting experiments. Based on the results of solving the second task using three different approaches, conclusions can be drawn about the significance of the results obtained, given that the models assess the importance of the influence of variable values in the context of their complex interaction with each other. The information obtained has potential value for experimenters in their optimal management of variable importance values and in organising experiments to create new materials with predicted values of the selected variable.

1 | Introduction

The influence of mechanical properties and heat resistance of epoxy composites on the durability of parts and resource saving of water transport means. Construction materials [1–3], especially epoxy composites, are promising for the transport industry

due to their high strength, corrosion and wear resistance, and ability to form complex shapes on the surface of large structures. At the same time, it should be noted that epoxy composite materials (CMs) have increased physical and mechanical properties and heat resistance, which directly affect the reliability and durability of important components and mechanisms of water

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transport, such as deadwood shafts and bearings, propellers, turbine blades, deck mechanisms and hull structures. The authors in [4–13] have proven that a reduction in destructive stresses, modulus of elasticity and impact toughness of epoxy CM by 15%–25% when the temperature rises to 70°C–90°C can lead to critical plastic deformation of protective coatings on parts and mechanisms that are operated under dynamic loads and seawater conditions. Thermal and corrosion–erosion ageing of these materials under conditions of prolonged operation accelerates the processes of underfilm corrosion, the formation of local cracks, and reduces resistance to fatigue creep. The authors in [14–28] have established that the introduction of modifiers, plasticisers and nanofillers into the epoxy oligomer in small quantities increases the adhesion and mechanical properties of epoxy CMs described above by 20%–35%. In addition, this also made it possible to increase the glass transition temperature of the composites by 15°C–20°C, which significantly improved their heat resistance. The above, in turn, made it possible to increase the overhaul intervals of vessels from 3–5 to 5–7 years, which accordingly reduced the operating costs of water transport. The authors in [29] examined the effect of aluminium oxide filler on the mechanical properties, water absorption, thermal degradation and wear of reinforced epoxy composites. NaOH treatment significantly improved the tensile strength, flexural strength and impact toughness of the composites by strengthening the bond between the fibre and the matrix. The study shows that NaOH treatment and the addition of aluminium oxide improve their mechanical, thermophysical and tribological properties. This makes them suitable for high-performance industrial applications. The authors in [30] optimised the composition of environmentally friendly polyester composites filled with Roselle fibres and boiled eggshells. The composition of ingredients in composites with increased wear resistance has been optimised. This study offers a new approach to optimising layer parameters in materials for practical applications, improving environmentally friendly manufacturing practices for CMs.

It is well known [21–25] that resource conservation in water transport directly depends on the ability of the CM to maintain properties under the influence of static, dynamic and thermal loads in the seawater environment. Epoxy composites, which are characterised by improved adhesion to a metal substrate due to the optimal combination of ingredients, are additionally characterised by resistance to water absorption (swelling), which prevents diffusion processes and the penetration of aggressive molecules to the surface of the substrate. As a result, the introduction of epoxy adhesives reduces the rate of corrosion and mechanical wear by 30%–50% [23]. In addition, the integration of epoxy composites into the construction of mechanisms (casting parts or products as a whole) reduces the total weight of the vessel, which ensures a 15%–20% reduction in fuel consumption and a reduction in emissions into the world's oceans. At the same time, it should be noted that today manufacturers are setting new requirements for improving the performance of water transport vehicles through the use of new protective coatings. This can be achieved by introducing into the epoxy binder not only new plasticisers and modifiers but also nano- and microfillers at a predetermined critical content. Thus, optimisation of the composition of protective coatings based on epoxy composites in terms of physical and mechanical properties and heat resistance is an important factor in improving the environmental friendliness and economic efficiency of water transport.

2 | Solving the Problem of Predicting Properties Using Various Machine Learning (ML) Algorithms

Traditional methods of designing the composition and, accordingly, the structural parameters of epoxy composites are based on establishing patterns in the dynamics of empirical dependencies of property indicators on additive content. This requires conducting a significant number of experiments, which entails high economic and energy costs and significantly slows down the development of new materials. ML offers a revolutionary approach, allowing the construction of prediction models based on pre-established data on the composition, structure and properties of composites. Works [29, 30] present a methodology of artificial neural networks (ANNs) and response surfaces that accurately predicted and optimised the wear behaviour of composites with aluminium oxide particles. To achieve this goal, an ensemble machine learning (EML) model was developed, including k-nearest neighbours (*k*-NNs), random forest, long short-term memory (LSTM) networks, and ANNs. The developed model was combined with the response surface methodology (RSM). This made it possible not only to optimise the composition of materials but also to establish the optimal parameters for layer thickness in composites. The authors in [31, 32] have established that regression algorithms, such as random forest and gradient boosting (GB), provide high accuracy in predicting the mechanical properties and structural characteristics of materials based on the physicochemical activity and geometric parameters of fillers and polymerisation modes. In particular, the authors in [33] substantiated the prospects of advanced modelling methods, including GB–optimisation of the number of particles, as well as in combination: GB–Bayesian optimisation and GB–differential evolution (GB–DE). Based on the prediction of CM strength, they proved that the GB–DE model is quite accurate with a value of $R^2 = 0.995,029$ and a root mean square error (RMSE) of 0.023862. In addition, an open-source graphical interface based on GB–DE is presented to provide engineers with a transparent tool for optimising concrete CM structures, eliminating the opacity of traditional ML models. This allows predicting the strength of composites with an error of $\leq 4\%$ using data on filler content, particle size and CM curing time. The authors in [34] used various methods to represent data on CM composition. At the same time, the XGBoost model was used to train and predict the thermal conductivity and mechanical properties of CM, thanks to which high accuracy was achieved with an average value of $R^2 = 0.95$. It has been proven that this machine learning model, based on experimental data, is a useful tool for predicting and optimising the properties of polymer composites.

In [32], research has demonstrated the potential of using ML to accelerate design, characterisation analysis and material composition optimisation. The authors used a combination of principal component analysis (PCA) and convolutional neural networks (CNNs) to predict the stress–strain curve and the final stage of composite failure (beyond the elastic limit). It has been shown that CNN transforms stress–strain curves into an effective latent space by visualising the PCA eigenbasis. Despite the availability of a dataset containing only 10%–27% of possible microstructure configurations, the average absolute prediction error is $< 10\%$ of the range of values in the dataset when measuring model performance based on material derivative descriptors such as modulus, strength and viscosity.

The authors in [35, 36] proposed a deep learning algorithm that allows synergistic combination of mathematical models and data. An effective mechanism for regularising the training of deep neural networks in low-data modes is proposed. The proposed methods and models allow scientific predictions and research to be carried out based on a small amount of data. In [37], it is proven that clustering methods, such as *k*-means, allow hidden patterns to be identified in large datasets by grouping materials with similar properties for further analysis. It can be used for both classification and regression tasks. Unlike other methods that build a general internal model, *k*-NN stores all data related to training data in *n*-dimensional space. In *k*-NN, the distances between new data and training data are in the hyperspace of descriptors. Thus, the result is predicted based on the values of the *k*-nearest components. The above properties of *k*-NN have attracted considerable interest among researchers in the field of predicting the mechanical behaviour of CM.

The authors in [31] integrated MD modelling methods to build ML prediction models. Initially, experiments were conducted to model the tensile properties of composites using MD modelling, and stress-strain curves were obtained from which the modulus of elasticity and tensile strength were determined. Next, the MD modelling data were fed into a feedforward neural network using support vector methods and AdaBoost regression algorithms. These models are trained to create an ML model capable of accurately predicting both the elastic modulus and the tensile strength. By comparing the predictive performance of different ML models, the most effective model for predicting the mechanical properties of CM under tension was determined.

The paper [38] demonstrates the feasibility of applying innovative models of deep CNNs and the mechanisms of structure formation and mechanical properties of CM. The models demonstrate the ability to distinguish and use complex relationships between the structure of materials and their mechanical properties. Through comprehensive experiments, the authors have verified that this approach achieves sufficient prediction accuracy at manageable training and computation costs.

Thus, ML not only speeds up the development process but also opens up opportunities for creating CM with predictable properties. In this context, ML becomes an indispensable tool for accelerated analysis of CM properties and optimisation of their composition.

3 | A Comprehensive Approach Combining ML and Mathematical Statistics Methods in Predicting the Properties of Epoxy Composites

The most promising direction is the combination of ML methods with classical mathematical statistics to create ensemble forecasting algorithms [39–47]. Ensemble algorithms, especially those with deep architecture, are capable of modelling complex nonlinear relationships between the composition of composites and their performance characteristics. Studies show [32, 39–42] that ensemble algorithms and CNNs can analyse microstructural images of composites and predict their strength with an accuracy of up to 92%. Recurrent neural networks (RNNs) are effective for modelling the behaviour of materials under cyclic loading, taking into account the history of deformations [32, 42]. The integration of statistical methods such as ANOVA and regression analysis

allows the statistical significance of input parameters to be determined and the reliability of predictions to be assessed. Hybrid models combining random forest [31] for feature selection and ANNs for regression demonstrate a 10%–15% reduction in RMSE compared with individual methods. An important aspect is the use of the PCA method [32, 46] to reduce the amount of data at the input of the neural network, which speeds up training and improves the interpretation of results. Such models allow not only to predict the properties of composites but also to generate new CM compositions with predetermined characteristics using the generation of new networks. In addition, the integration of physics-informed neural networks [39–47] ensures that the prediction complies with the fundamental laws of mechanics and thermodynamics. This comprehensive approach opens up new opportunities for creating digital twins of CMs, allowing their behaviour to be modelled in real operating conditions without conducting costly and time-consuming experiments. Thus, the comprehensive use of ML algorithms and statistical methods opens up new horizons for the development of new-generation epoxy composites.

4 | Materials and Methods of Research

During the formation of polymer compositions, DER 331 epoxy resin (ISO 9001), which is a product of the reaction of epichlorohydrin and bisphenol A, was selected as the base. In order to improve the rheological properties of the epoxy binder, aliphatic resin DEG-1 (TU 2225–390-04872688–98) was added to the DER 331 resin as a plasticiser. DEG-1 is added to epoxy resins as a plasticiser with active diluent properties, giving the resin elasticity, impact strength and resistance to temperature changes. DEG-1 resin combines well with epoxy resin. Recommended concentrations are 10%–12%. To further improve the performance characteristics of the developed polymer composites and coatings based on them, an active modifier was used—phthalic anhydride or 2-benzofuran-1,3-dione. Its content in the DER-331 epoxy oligomer was varied within the range of 0.1–3%. Polyethylene polyamine (PEPA) (TU 2413–357-00203,447–99) was used to cure the epoxy compositions. The chemical formula of PEPA is as follows: $H_2N(CH_2CH_2NH)_nH$. It should be noted that PEPA is a widely used and economical hardener for epoxy resins.

Nano- and micro-sized powders are used as fillers to improve the physical and mechanical properties and heat resistance of CM. In particular, nanotubes in the form of MWCNT powder developed by Guangzhou Hongwu Material Technology Co., Ltd. (China) are used. MWCNT nanopowder is nanoclass graphite crystals, mono- or multilayered flaky graphite in the form of MWCNT. The work uses multilayered nanotubes with the following parameters: diameter: 10–30 nm and length: 5–20 μ m. Nanosilver particles with a dispersion of 10–100 nm were also used as a nanoadditive. Nanosilver is resistant to any acidic and neutral environments in the absence of oxygen. Nanosilver provides high antibacterial activity against microorganisms.

Oxytetracycline powder was used as a microadditive. Oxytetracycline ($C_{22}H_{24}N_2O_9$) is a tetracycline antibiotic used to combat bacterial organisms. The additive with a particle size of 5–10 μ m can improve the cohesive strength of CM, which led to its selection in the formation of corrosion-resistant epoxy protective coatings for ship hulls.

The study investigated the following properties of CM: destructive stress and modulus of elasticity in bending, impact strength, adhesive strength, residual stress and heat resistance.

The tensile strength and modulus of elasticity in bending were determined using a four-point loading scheme in accordance with ASTM D 790-03. The parameters of the samples for isotropic materials were length $l = 120 \pm 2$ mm, width $b = 15 \pm 0.5$ mm and height $h = 10 \pm 0.5$ mm. The tensile strength values were determined using the following formula:

$$\sigma_3 = \frac{3F_{\max}L}{2bh^2}, \quad (1)$$

where $\Delta F_{\max}$ is the maximum load before sample failure, N; L is the distance between the supports, mm; b is the width, mm; and h is the thickness, mm.

The elastic modulus was determined according to the following formula:

$$E = \frac{0,185 \cdot L_v^3 \cdot \Delta F}{bh^3 \cdot \Delta z}, \quad (2)$$

where L_v is the distance between supports, mm; ΔF is the load, N; b is the width, mm; h is the thickness, mm; and Δz is the deflection of the sample, mm.

The impact strength of KM was studied using the Charpy method without notching (ISO 179-1) on a MK-30 pendulum impact tester at a temperature of $T = 298 \pm 2$ K and relative humidity of $\varphi = 50 \pm 5\%$. Sample dimensions: $(65 \times 12 \times 12) \pm 0.5$ mm. Distance between supports: 40 ± 0.5 mm. The impact strength of CM was determined using the following formula:

$$W = \frac{A_n}{bs} 10^3, \quad (3)$$

where A_n is the impact energy consumed to fracture the specimens without cuts; b is the width, mm; and s is the thickness, mm.

Adhesive strength at peel was tested in accordance with ISO 4624:2016 'Paints and varnishes. Determination of adhesion by peel'. The aim was to determine the adhesive strength between the coating and the substrate. The test method consisted of gluing pistons with a diameter of $d = 25 \pm 0.1$ mm with the test composition and polymerising them. A testing machine was used to create a perpendicular peel force and record the maximum stress (in MPa) at which failure occurred. It should be noted that the loading rate must be constant (1.0–1.5 MPa/s), and the environmental conditions must be as follows: temperature $23^\circ\text{C} \pm 2^\circ\text{C}$ and relative humidity $50 \pm 5\%$.

Additionally, residual stresses in coatings after their polymerisation were analysed. The method consists in measuring the stresses that arise in the polymer coating during its compaction (bonding) on a metal substrate. The essence of the method: a metal substrate (0.18-mm thick) was fixed as a console (one end was rigidly fixed and the other was free). A thin layer of adhesive (0.1–0.2 mm) was applied to the substrate. During polymerisation, the adhesive shrunk. This resulted in stresses that bent the metal substrate. After determining the amount of deflection of the free end of the substrate and taking into account the modulus of elasticity of the base, the geometric parameters of the substrate and the adhesive, the residual stresses in the polymer were

calculated using the formula [16, 18]. Thus, using a simple technique and an uncomplicated mechanical model, the residual stresses were quantitatively evaluated without complex equipment.

The heat resistance of the composites was analysed in accordance with ISO 75-2. In this case, the parameters of the temperature index were determined based on the bending values of the samples at $h = 6$ mm [23, 24].

For each composition option, five identical samples ($n = 5$) were manufactured and tested in accordance with the requirements of the above standards: ASTM D 790-03, ISO 179-1, ISO 4624:2016 and ISO 75-2.

The replication strategy in this study was developed based on three key principles: statistical power, technical standards and efficient resource allocation. In particular, a statistical justification for the number of replicates ($n = 5$) was applied. This number is the minimum sufficient for applying parametric statistical tests and calculating reliable confidence intervals. Assuming a standard deviation of 7%–10%, typical for composites, five replicates allow detecting a statistically significant difference between series at the level of 15%–20% with a probability (test power) $> 80\%$ at a significance level of $p < 0.05$. This corresponds to acceptable standards in materials science. Additionally, a quality control and outlier detection strategy was applied. In addition to quantity, the proposed strategy includes a data processing protocol using a hierarchical replication structure that corresponds to the experimental design. Not only the measurements were replicated but also the sample manufacturing process. Each composition was synthesised in five independent batches (replication at the synthesis level). Five samples were produced from each batch for testing (replication at the testing level). This two-level strategy allows for the statistical separation of sources of variability: synthesis error from measurement error, providing a much deeper understanding of the reproducibility of the technological process. This comprehensive strategy, combining standard requirements, statistical principles and practical constraints, ensures scientific rigour, allows valid conclusions to be drawn and guarantees that the results obtained are reliable and reproducible in other laboratories.

5 | Technology and Temperature-Time Parameters for Forming Polymer Composites

The technology for forming polymer composites is as follows. Before forming compositions from independent ingredients, preparatory work should be carried out since after combining the components, the gelatinisation period of the resulting mixtures is 40–60 min. The components are mixed manually for 4–6 min using a drill with a special mixer at a rotor speed of $600\text{--}800 \text{ min}^{-1}$. In addition, in some cases, if possible, the ingredients are combined hydrodynamically at a temperature of $25^\circ\text{C}\text{--}30^\circ\text{C}$ for 10 ± 0.1 min, followed by ultrasonic treatment (UST) of the compositions for 1.5 ± 0.1 min. After cooling the mixture to room temperature (60 ± 5 min), a hardener is added and the compositions are mixed for 5 ± 0.1 min.

The mould should be filled slowly and continuously to allow air pockets to be removed from the composition and to avoid trapping new air pockets. Pouring should be carried out continuously until the mould is completely filled, after which the

material is kept at room temperature (295 ± 2 K) for 12.0 ± 0.1 h. The next step is to heat the composition to a temperature of 393 ± 2 K, hold it at this temperature for 2.0 ± 0.05 h and cool it to a temperature of 293 ± 2 K.

6 | Results of Experiments

Based on the results of the experiments, the properties of epoxy CMs were obtained, which are presented in Tables 1, 2, 3 and 4. Note that Tables 1, 2 and 3 were used only for ML (training dataset). CM based on DER-331 epoxy resin (100%), TETA hardener (10%), phthalic anhydride modifier and fillers with different contents: nanosilver and oxytetracycline were used. Table 4 uses CM based on DER-331 resin (100%), DEG-1 plasticiser (12%), PEA hardener (12%) and nanotube filler with different powder contents (0.05% and 0.1%). Note that Table 4 uses only data obtained for CM without powder (Filler – 0%) for ML, as well as all indicators of CM properties with filler, except for heat resistance. Heat resistance indicators are the subject of prediction using various ML algorithms.

Thus, the models were trained on the relationship between the composition of CM and mechanical/physical properties, where heat resistance was known (Tables 1, 2 and 3) or hidden only for target samples (Table 4).

A target subset of data was selected for testing and validating the predictive ability of the models: heat resistance indicators for CM with filler (0.05% and 0.1% nanotubes) from Table 4. These data were never fed into the models during training and were used exclusively to assess the accuracy of the main target variable prediction. For internal verification of the stability of the models on the training data (Tables 1, 2, 3 and part of Table 4), 5-fold cross-validation was used.

This strategy simulates a real design task: a model trained on a large data set from different systems must predict a critical operational characteristic (heat resistance) for a new but similar hybrid system (with a different plasticiser and hardener) using only the known mechanical and physical properties of these materials. This ensured complete transparency and allowed us to evaluate the real predictive power of the proposed ML algorithms.

For a better interpretation of the research process, we provide a block diagram illustrating the integration of experimental stages with the ML pipeline, including data generation, modelling and prediction validation. In particular, the flowchart of the experimental and computational pipeline (Figure 1) includes the following key stages:

1. Experimental module: CM formation, testing (mechanical/thermophysical properties) and data table formation.
2. Preliminary data processing: normalisation, division into training/test samples and feature selection.
3. ML modelling: application of ensemble algorithms and cross-validation.
4. Validation and prediction: evaluation of models on independent data and prediction of composite heat resistance.
5. Interpretation of results: analysis of feature importance and conclusions for material design.

7 | Modelling Dependency

Based on the experimental data presented above (Tables 1, 2, 3 and 4), this paper considers the problem of predicting the values of the variable heat resistance depending on the values of the following CM properties: filler, destruction stresses, elastic modulus (elasticity), impact strength (viscosity), adhesive strength (solidity) and residual stresses.

The programme implementation was carried out in Visual Studio Code using Jupyter Notebook documents and the scikit-learn v1.7.2 library [48].

Various ML algorithms were used to solve the forecasting problem: AdaBoost (ada_boost), bagging (bagging), decision trees (decision_trees), gradient boost (gradient_boost), random forest (random_forest), stacking (stacking) and voting (voting), as well as basic models for comparison: ridge regression (ridge), support vector machine (SVM) and k -NN, which establish the dependence of the values of the variable heat resistance on the values of the variables: filler, destruction stresses, elastic modulus, impact strength, adhesive strength and residual stresses. In fact, these algorithms are different models of such a dependence. To ensure the reliability of the results and minimise dependence on the specifics of individual algorithms, a comprehensive approach combining ML and mathematical statistics methods was used.

Thus, the following models were considered for forecasting:

$$y = f_j(x_0, x_1, x_2, x_3, x_4, x_5), \quad j = \overline{0, 9}, \quad (4)$$

where y is the heat resistance variable; f_0 is the AdaBoost machine learning algorithm (ada_boost); f_1 is the bagging machine learning algorithm (bagging); f_2 is the decision trees ML algorithm (decision_trees); f_3 is the gradient boost machine learning algorithm (gradient_boost); f_4 is the random forest ML algorithm (random_forest); f_5 is the machine learning algorithm stacking (stacking); f_6 is the ML algorithm voting (voting); f_7 is the model ridge regression (ridge); f_8 is the model SVM; f_9 is the model k -NN; x_0 is the variable filler; x_1 is the variable destruction stresses; x_2 is the variable elastic modulus (elasticity); x_3 is the variable impact strength (viscosity); x_4 is the variable adhesive strength (solidity) and x_5 is the variable residual stresses.

Given the volume and structure of the experimental data ($N = 115$), the number of base estimators in the ensembles was limited to $n_{\text{estimators}} = 5$. This choice is a compromise between the need to create diverse models in the ensemble and minimising the risk of overfitting due to excessive overlap of bootstrap samples. The parameters of the base estimators were left at their default values in order to control model complexity and preserve interpretability. The effectiveness of this configuration was empirically confirmed by high and stable accuracy within the applied evaluation criteria.

The models (ML algorithms) used in the work had given values of their parameters. Model f_0 (ML algorithm AdaBoost) was used with the parameter values: $n_{\text{estimators}} = 5$, $\text{loss} = \text{'linear'}$, $\text{random_state} = 42$, other parameters are default. Model f_1 (ML algorithm bagging) was used with the parameter values: $n_{\text{estimators}} = 5$, $\text{bootstrap} = \text{false}$, $\text{random_state} = 42$, other parameters are default. Model f_2 (ML algorithm decision trees) was used with the parameter values: $\text{random_state} = 42$, other

TABLE 1 | Properties of CM based on DER-331 resin (100%), TETA hardener (10%) and nanosilver filler (powder content q , %).

Filler (%)	Destruction stresses (MPa)	Elastic modulus (GPa)	Impact strength (kJ/m ²)	Adhesive strength (MPa)	Residual stresses (MPa)	Heat resistance (K)
0	79.8	2.8	9.6	9.9	0.94	360
	88.9	3.2	4.9	17.1	0.94	345
	80.4	2.8	6.4	10.4	0.98	356
	115.2	2.5	9.6	11.6	1.82	362
	91.5	3.4	22.4	9.9	1.48	358
0.025	61.7	2.5	8.8	11.2	1.24	361
	88.5	3.2	12.9	9.8	0.93	359
	84.6	3.4	10.9	6.2	0.76	343
	70.3	2.8	19.3	14.6	1.22	361
	81.1	3.1	14.1	14.6	0.99	356
0.05	86.5	3.3	13.8	12.9	1.06	357
	107.5	3.6	17.6	21.2	1.38	360
	88.2	2.8	14.5	14.6	0.98	357
	87.8	3.4	20.7	23.9	0.71	357
	96.8	3.4	17.5	12.7	1.22	359
0.075	94.5	3.3	16.9	17.1	1.02	359
	101.4	3.1	20.9	19.8	0.69	356
	83.2	3.0	13.2	14.1	1.24	354
	103.3	3.4	10.9	17.1	0.79	361
	95.9	3.2	15.0	12.9	0.88	357
0.1	102.3	3.3	16.4	29.5	0.96	355
	104.5	3.4	18.8	29.1	1.30	355
	83.4	2.9	13.9	13.9	0.82	355
	76.1	3.4	8.0	27.3	1.64	355
	88.0	3.2	13.6	14.4	1.27	356
0.125	93.9	3.2	14.2	29.1	1.38	355
	99.0	3.0	12.2	13.9	0.41	359
	80.4	3.0	13.4	11.4	1.28	357
	113.2	3.4	11.8	11.7	0.50	356
	97.3	3.2	12.4	13.5	0.76	358
	94.2	3.2	13.1	10.4	0.88	357

TABLE 2 | Properties of cements based on DER-331 resin (100%), TETA hardener (10%) and phthalic anhydride modifier (additive content q , %).

Filler (%)	Destruction stresses (MPa)	Elastic modulus (GPa)	Impact strength (kJ/m ²)	Adhesive strength (MPa)	Residual stresses (MPa)	Heat resistance (K)
0	79.8	2.8	9.6	9.9	0.94	360
	88.9	3.2	4.9	17.1	0.94	345
	80.4	2.8	6.4	10.4	0.98	356
	115.2	2.5	9.6	11.6	1.82	362
	91.5	3.4	22.4	9.9	1.48	358
0.1	61.7	2.5	8.8	11.2	1.24	361
	86.6	3.2	25.2	13.5	1.03	363
	77.6	2.8	10.3	15.4	0.97	361
	92.9	2.7	9.0	17.9	0.57	358
	85.8	2.9	15.2	17.3	0.83	360
0.25	86.4	3.0	14.4	16.4	0.94	362
	80.1	2.9	12.9	21.2	0.85	362
	81.9	3.1	10.9	26.8	0.45	359
	61.1	2.8	3.3	13.5	0.95	347
	75.6	2.9	9.2	26.0	0.88	360
0.5	78.1	3.0	10.1	22.3	0.74	362
	82.2	3.2	6.6	21.0	0.63	362
	87.6	3.5	4.9	21.0	0.96	353
	69.0	2.8	4.7	14.1	0.86	361
	78.1	3.1	5.5	21.2	0.85	360
1	80.2	3.2	5.2	19.8	0.74	359
	28.3	2.2	6.2	15.0	0.77	353
	96.7	3.0	3.9	13.7	0.73	358
	73.0	3.4	5.9	11.4	0.73	358
	66.0	2.9	5.3	17.9	0.79	356
2	84.9	3.2	5.5	14.7	0.81	357
	66.8	4.0	4.6	14.4	0.85	368
	24.8	3.2	5.5	16.2	0.86	356
	56.4	2.8	4.7	16.0	0.63	359
	49.3	3.3	4.8	14.6	0.78	362
3	61.5	3.4	5.1	15.5	0.71	361
	78.5	3.4	4.7	15.0	0.75	359
	19.8	1.9	4.5	24.3	0.51	360
	70.4	3.1	4.6	15.2	0.50	356
	56.2	2.8	4.6	28.7	0.58	358
	74.5	3.3	4.5	22.3	0.60	359

TABLE 3 | Properties of cements based on DER-331 resin (100%), TETA hardener (10%) and oxytetracycline filler (powder content q , %).

Filler (%)	Destruction stresses (MPa)	Elastic modulus (GPa)	Impact strength (kJ/m ²)	Adhesive strength (MPa)	Residual stresses (MPa)	Heat resistance (K)
0	79.8	2.8	9.6	9.9	0.94	360
	88.9	3.2	4.9	17.1	0.94	345
	80.4	2.8	6.4	10.4	0.98	356
	115.2	2.5	9.6	11.6	1.82	362
	91.5	3.4	22.4	9.9	1.48	358
0.5	61.7	2.5	8.8	11.2	1.24	361
	62.5	3.4	5.6	19.1	1.20	349
	76.7	3.1	4.1	42.8	0.80	362
	49.1	3.0	14.2	36.8	1.38	361
	62.7	3.2	8.3	28.5	1.24	359
1.0	69.6	3.3	7.8	32.0	1.18	360
	47.3	3.1	3.9	16.6	1.14	339
	82.6	2.9	5.6	21.8	1.00	362
	54.7	2.9	5.7	17.7	1.24	359
	64.3	3.0	5.5	21.6	1.12	358
1.5	66.1	3.1	5.8	19.8	1.18	360
	49.3	3.3	5.9	28.5	1.33	357
	51.4	3.5	4.0	23.9	0.90	359
	58.0	3.5	10.1	34.9	0.84	362
	53.1	3.4	6.3	29.2	0.95	359
2	54.0	3.5	7.4	30.1	1.01	360
	58.0	3.1	6.0	33.1	1.49	355
	62.3	3.1	9.0	42.4	1.22	355
	49.5	2.7	6.9	28.9	0.97	352
	59.4	2.9	7.2	30.0	1.13	354
2.5	57.2	3.0	7.8	34.2	1.28	354
	67.7	2.8	8.0	28.5	1.63	355
	51.7	2.9	9.5	30.6	1.13	361
	52.5	2.6	5.7	41.2	1.26	359
	57.8	2.7	7.8	35.2	1.28	358
3	56.3	2.8	8.3	34.0	1.36	358
	55.7	2.7	3.2	26.8	0.97	360
	45.7	3.2	4.7	22.1	0.91	355
	50.7	2.9	4.6	31.8	1.24	357
	51.2	3.1	4.4	32.5	1.03	358
	50.1	3.0	4.2	29.4	1.12	357

TABLE 4 | Properties of CM based on DER-331 resin (100%), DEG-1 plasticiser (12%), PEPA hardener (12%) and nanotube filler (powder content q , %).

Filler (%)	Destruction stresses (MPa)	Elastic modulus (GPa)	Impact strength (kJ/m ²)	Adhesive strength (MPa)	Residual stresses (MPa)	Heat resistance (K)
0	51.6	2.2	9.6	29.9	1.42	356
	60.9	3.2	6.9	36.1	1.66	348
	50.4	1.8	6.7	30.5	0.95	356
	85.2	2.5	13.8	38.6	1.86	344
	61.5	3.4	14.7	29.8	0.48	351
0.05	97.2	2.6	16.4	61.2	1.22	353
	68.0	2.8	14.2	54.6	0.99	358
	67.6	3.4	10.7	43.9	0.81	353
	75.8	2.9	12.8	43.5	1.22	360
	78.0	3.2	13.9	47.1	0.48	354
0.1	84.0	3.2	8.8	39.4	0.48	356
	73.3	2.7	12.5	44.8	0.83	355
	66.4	3.5	10.1	38.7	1.03	352
	78.1	3.4	12.4	48.1	0.81	351
	60.3	3.6	12.2	42.5	0.68	350

parameters are default. Model f_3 (ML algorithm gradient boost) was used with the parameter values: $n_estimators = 5$, $random_state = 42$, other parameters are default. Model f_4 (ML algorithm random forest) was used with the parameter values: $n_estimators = 5$, $bootstrap = false$, $random_state = 42$, other parameters are default. Model f_5 (ML algorithm stacking) was used with the parameter values: estimators (linear regression ()): all parameters are default, decision tree regressor ($random_state = 42$), random forest regressor ($n_estimators = 5$, $random_state = 42$)), other parameters are default. Model f_6 (ML algorithm voting) was used with the parameter values: estimators (random forest regressor ($n_estimators = 5$, $random_state = 42$), gradient boost regressor ($n_estimators = 5$, $random_state = 42$), linear regression ()):all parameters are default), weights = [1.3, 1.3, 0.9], other parameters are default.

The software implementation of all models (ML algorithms) was performed using the scikit-learn v1.7.2 library [48].

To train and test the models, we generated dataset X . This dataset X was obtained by combining the data in Tables 1, 2, 3 and 4 by rows. However, in order to preserve information about the materials of these data tables, an additional variable 'Material type' was introduced with values: '1' for Table 1, '2' for Table 2, '3' for Table 3 and '4' for Table 4. As a result, Formula (4) for models $f_j, j = \overline{0, 9}$, took the following form:

$$y = f_j(x_0, x_1, x_2, x_3, x_4, x_5, x_6), \quad j = \overline{0, 9}, \quad (5)$$

where x_6 is the variable Material type; definitions of other variables are described in (4).

After that, the dataset X was preprocessed by removing rows that contained missing data in at least one column. The next step was to form a training X_{train} and a test X_{test} dataset from the obtained dataset X . The training sample X_{train} included the rows of the dataset X that were in Tables 1, 2 and 3, as well as the rows from Table 4 for which the value of the variable Filler = 0%. And the test sample X_{test} includes the rows of the dataset X that were in Table 4 with the values of the variable filler = 0.05% and filler = 0.1%.

7.1 | Model Optimisation and Training

In order to obtain the optimal parameters for models $f_0 \dots f_9$, GridSearchCV from the sklearn library, model_selection module [49] was used. For the correct selection of hyperparameters, the training sample X_{train} was additionally divided into two subsamples: CV-Train (80% of X_{train}) and CV-Val (20% of X_{train}). The CV-Train subsample was used directly for training models in the process of searching for hyperparameters, while the CV-Val subsample was used to evaluate the quality of models for each hyperparameter configuration. This division ensures the independence of the evaluation from the training data and avoids information leakage when selecting parameters. The procedure is repeated k times so that each fold acts as a validation fold exactly once. The final quality assessment for a given hyperparameter configuration is calculated as the average metric value across all k iterations.

The choice of $k = 3$ is due to the limited size of the training sample ($N_{train} = 108$): a smaller number of folds provides a sufficient size of internal validation subsamples (~36 samples) for stable metric evaluation. To verify the stability of the results,

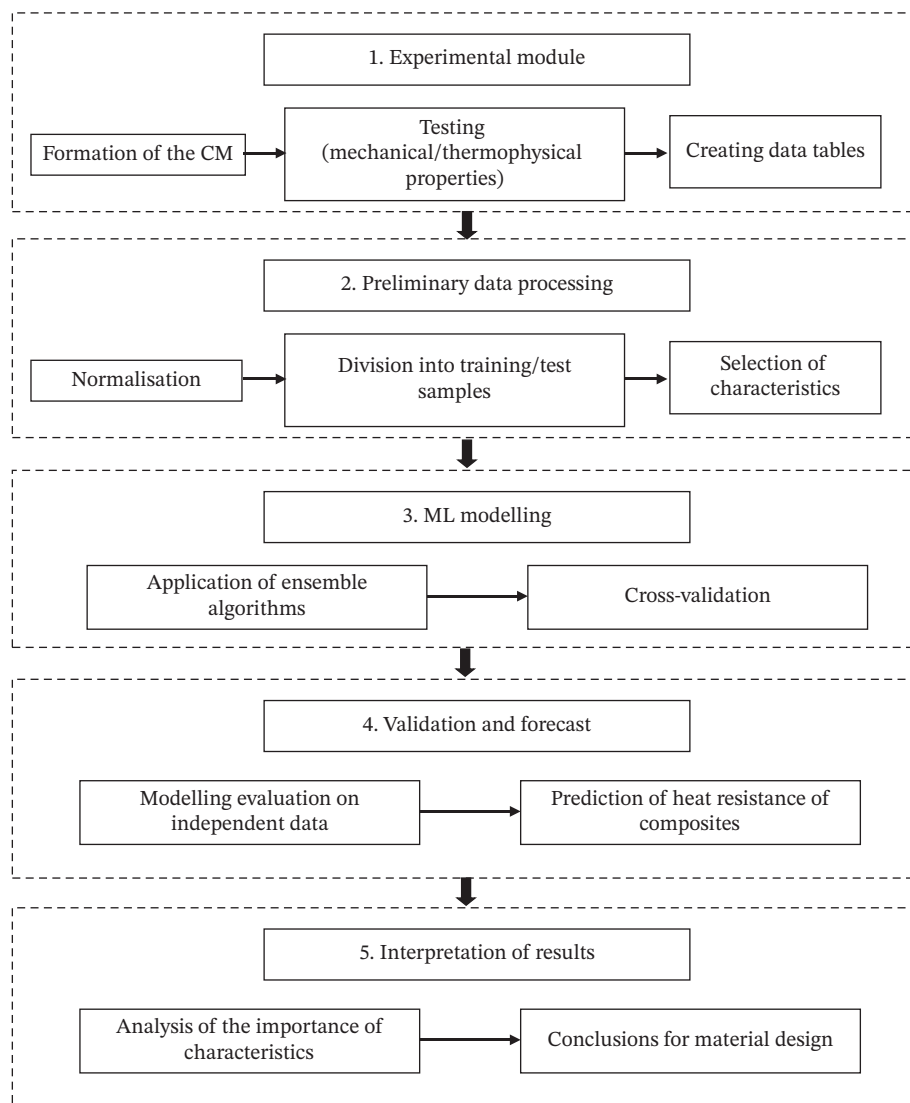


FIGURE 1 | Block diagram of the experimental computing pipeline.

additional validation was performed with $k = 5$, which showed almost identical results: the best model achieved a MAPE value of 0.685% ($k = 5$) versus 0.698% ($k = 3$), the worst: 1.176% ($k = 5$) versus 1.454% ($k = 3$). Such consistency indicates that the choice of k is not a critical factor for this task.

After determining the best configuration of hyperparameters, the final model was retrained on the full training sample X_{train} , which is referred to as Final-Train. The Final-Train subsample coincides with X_{train} and includes rows from the dataset in Tables 1, 2 and 3, as well as rows from Table 4 for which the value of the filler variable is 0%. The quality of the final model was evaluated on an independent test sample X_{test} (Final-Val), which included rows from Table 4 with values of the variable filler = 0.05% and filler = 0.1%. The use of an independent test sample Final-Val, which has never been involved in training or hyperparameter selection, ensures an unbiased assessment of the generalisation ability of the constructed models.

The following quality metrics were used for quantitative comparison of models: mean squared error (MSE), mean absolute percentage error (MAPE), mean absolute error (MAE) and coefficient of determination (R^2). The MSE, MAE and MAPE metrics characterise

the absolute and relative values of deviations of predicted values from actual values, while R^2 reflects the proportion of variance in the target variable explained by the model and serves as a generalised indicator of approximation quality.

7.2 | Learning Outcomes

Metrics according to the results of forecasting the values of the variable heat resistance depending on the values of the variables: filler, destruction stresses, elastic modulus, impact strength, adhesive strength, residual stresses and material type using models f_j , $j = 0, 9$, on different samples are given in Tables 5, 6, 7 and 8.

Analysis of the results of models with optimised hyperparameters revealed that random forest (0.74%), bagging (0.74%) and ridge (0.75%) models showed the best results on the test sample, confirming the feasibility of comparing ensemble methods with simple base models.

During 3-fold cross-validation, voting (val MAPE 0.60%) and stacking (val MAPE 0.63%) showed the best results, indicating their ability to generalise well within the training data distribution. k -NNNN showed perfect accuracy in training (train

TABLE 5 | MSE metric values for different samples during cross-validation and training on optimal parameters.

Model	CV-Train	CV-Val	Final-Train	Final-Val
bagging	7.1267	20.4145	7.7472	12.656
decision_trees	13.9876	15.0376	14.4345	19.0
random_forest	3.7725	12.7463	4.3004	9.7835
gradient_boost	7.5221	13.5923	8.5375	12.249
stacking	5.2171	12.0904	10.7788	41.6985
ada_boost	4.8535	14.3495	6.0683	22.2593
voting	1.5509	9.7225	1.5667	15.2406
SVM	24.1658	12.6629	21.8995	23.0906
k-NN	0.0	13.8199	0.0	13.121
ridge	20.2112	16.7075	19.3061	9.0955

TABLE 6 | MAPE metric values for different samples during cross-validation and training on optimal parameters.

Model	CV-Train	CV-Val	Final-Train	Final-Val
bagging	0.5356	0.9432	0.5259	0.7429
decision_trees	0.7491	0.857	0.7452	0.9537
random_forest	0.3946	0.6425	0.4211	0.7377
gradient_boost	0.6314	0.7231	0.6388	0.7523
stacking	0.416	0.6266	0.5345	1.6534
ada_boost	0.4869	0.7082	0.5826	1.1958
voting	0.2606	0.5991	0.2651	0.8167
SVM	0.9576	0.6316	0.8903	1.1884
k-NN	0.0	0.666	0.0	0.8979
ridge	0.947	0.8449	0.9031	0.7517

TABLE 7 | MAE metric values for different samples during cross-validation and training on optimal parameters.

Model	CV-Train	CV-Val	Final-Train	Final-Val
bagging	1.9	3.3636	1.8759	2.64
decision_trees	2.6568	3.0589	2.6418	3.4
random_forest	1.4033	2.285	1.4956	2.6133
gradient_boost	2.2447	2.577	2.2704	2.6788
stacking	1.4726	2.2278	1.8868	5.8331
ada_boost	1.7378	2.5139	2.0809	4.2215
voting	0.926	2.1373	0.9417	2.9102
SVM	3.3833	2.2435	3.1477	4.1909
k-NN	0.0	2.3687	0.0	3.171
ridge	3.3617	3.0146	3.205	2.6607

MAPE 0.00%), which is a typical sign of data memorisation for neighbour-based methods.

However, the results on the test sample (Material 4, new filler concentrations) were significantly different. The models that showed the best results during cross-validation significantly deteriorated in quality when extrapolated:

- Stacking: 0.63% (CV) → 1.65% (Test) — 2.6 times worse
- Voting: 0.60% (CV) → 0.82% (Test) — 1.4 times worse
- AdaBoost: 0.71% (CV) → 1.20% (Test) — 1.7 times worse

Models with moderate cross-validation results proved to be more stable:

TABLE 8 | R^2 metric values for different samples during cross-validation and training on optimal parameters.

Model	CV-Train	CV-Val	Final-Train	Final-Val
bagging	0.6915	-0.6227	0.6358	-0.4447
decision_trees	0.3944	-0.1953	0.3215	-1.1689
random_forest	0.8367	-0.0132	0.7979	-0.1168
gradient_boost	0.6743	-0.0804	0.5987	-0.3983
stacking	0.7741	0.039	0.4933	-3.7601
ada_boost	0.7899	-0.1406	0.7147	-1.541
voting	0.9329	0.2272	0.9264	-0.7398
SVM	-0.0462	-0.0065	-0.0294	-1.6359
k-NN	1.0	-0.0985	1.0	-0.4978
ridge	0.125	-0.328	0.0925	-0.0383

- Random forest: 0.64% (CV) $\rightarrow$ 0.74% (Test) — only 15% worse
- Bagging: 0.94% (CV) $\rightarrow$ 0.74% (Test) — improvement by 21%
- Ridge: 0.84% (CV) $\rightarrow$ 0.75% (Test) — deterioration by 11%

For each model, learning curves were constructed that reflect the dependence of MAE on the size of the training sample. This allows us to evaluate the difference between the error on the training and validation samples, the rate of error reduction with increasing data, and signs of overfitting, namely, situations where train MAE tends to zero while validation MAE remains high.

A comparison of errors on the training and test samples revealed significant overfitting for most of the models shown in Figure 2.

A common feature of most of the models considered is a significant gap between Train MAE and Validation MAE for small training samples, which gradually narrows as the number of training samples increases. This behaviour is a sign of overfitting and indicates that with a limited amount of data, models memorise the training sample instead of generalising patterns.

The most pronounced overfitting is observed in ensemble models with a tree-like base. In particular, for decision trees, AdaBoost and gradient boost models, the training error remains close to zero across the entire range of sample sizes while the validation error is significantly higher and shows a wide confidence interval. This indicates that these models are overly complex and prone to memorising the training data. A similar, albeit slightly less pronounced, picture is demonstrated by the bagging, random forest and voting models: MAE on the training sample tends to zero, while MAE on the validation sample, despite a downward trend, remains at 15–25 units for the full training sample.

The SVM and ridge models demonstrate excellent performance. For SVM, the training and validation curves practically coincide across the entire range of sample sizes and stabilise at ~ 20 MAE units, indicating no overfitting and satisfactory generalisation ability of the model. The ridge model also shows no signs of overfitting in the classical sense: both curves are close, although they demonstrate some instability at small sample sizes.

The negative R^2 values on the test sample for all models (from -0.04 to -3.76) were significant. This means that the models perform worse than the simple average heat resistance value, which is typical for extrapolation tasks when the test data are outside the training range, especially for a small amount of validation data with low dispersion. Ridge showed the best R^2 (-0.04), while stacking showed the worst (-3.76).

7.3 | Assessment of the Importance of Characteristics

For each of the models $f_0, \dots, f_9$, two types of feature importance metrics were calculated. For tree-based models (decision trees, random forest, gradient boost and AdaBoost), Build In Importance was used, thanks to the built-in feature_importances_ attribute, which calculates the normalised sum of the impurity decrease. This method is fast and effective, but it has limitations: it can be biased towards high-cardinality features and does not take into account interactions between predictors. Taking into account the bias and the absence of the feature_importances_ attribute in the base models, it was decided to use the model-agnostic Permutation Importance method.

The algorithm works as follows: the base model quality metric is calculated on a given sample, for each feature x_i the values are randomly rearranged, and a new quality metric is calculated with the damaged feature. The final importance is calculated by (basic metric)-(metric after permutation), and the procedure is repeated 5 times to stabilise the results.

It is important to note that Permutation Importance was calculated separately for the training (X_{train}) and test (X_{test}) samples. This allowed us to identify features with divergent importance: those that are useful for training but harmful (or uninformative) for extrapolation.

Feature importance analysis was performed based on negative importance in the test sample. A feature is considered harmful for extrapolation if its average permutation importance among all models is negative, i.e., removing this feature improves the quality of the forecast. This indicates that the model uses dependencies that are not transferable to new data. The Feature Importance results are shown in Table 9.

In the training sample, residual stresses demonstrates moderate importance, but in the test sample, negative importance was found in 8 out of 9 models shown in Figure 3. The average



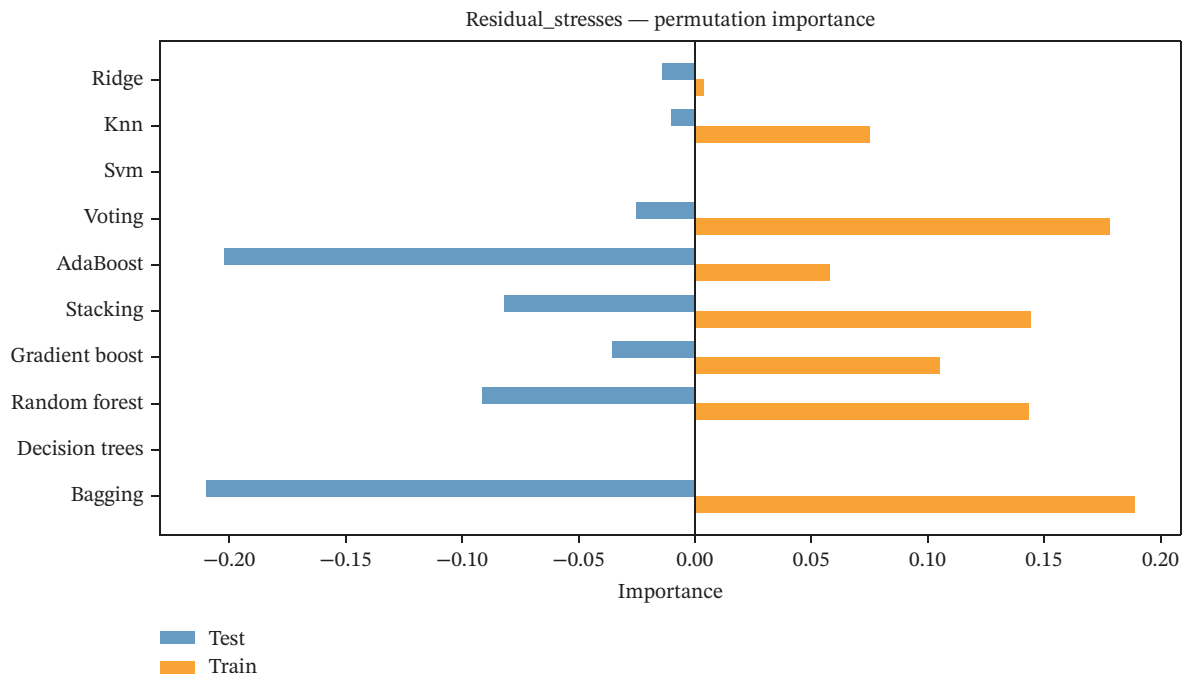
FIGURE 2 | Training curves for training and validation samples.

importance in the test sample is -0.067 . Residual stresses effectively distinguish Materials 1–3, but this relationship does not carry over to Material 4 at new filler concentrations. The models

captured specific patterns of residual stresses for each material type, and this information degrades the quality of the prediction when extrapolated.

TABLE 9 | Feature Permutation Importance among models.

Model	Filler	Destruction stresses	Elasticity	Viscosity	Solidity	Residual stresses	Material type
Bagging	0	0.113	−0.111	0.001	0	−0.209	0
decision_trees	0	0	0	0	0	0	0
random_forest	−0.001	0.018	0.12	−0.007	0	−0.09	0
gradient_boost	0	0.003	0	0.009	0	−0.035	0
Stacking	−0.001	0.008	0.14	0.028	0	−0.081	0
ada_boost	−0.006	−0.005	0.04	0.066	0	−0.202	0
Voting	−0.005	−0.07	0.024	−0.007	0.001255	−0.0251	0
SVM	−5.6E-07	0.0004	−5.13937E-07	0.0001	0.000483	−5.7E-06	0
k-NN	−0.00104	0.135	−0.0002	0.038	0.01	−0.01	0
Ridge	−0.005	−0.0176	0.0785	0.0112	0.0734	−0.0142	0
MEAN	−0.002	0.01848	0.0291	0.0139	0.0085	−0.066	0

**FIGURE 3** | Residual stresses importance among models.

Material type: average importance on the test = 0.000 (as expected, since the test only contains Material 4).

Filler: average importance on the test sample = −0.002 (practically zero, slightly negative).

Solidity: average importance on the test sample = +0.0085 (almost zero, only Ridge shows +0.073 and KNN +0.010).

Viscosity: average importance on the test sample = +0.0139 (low, although there are positive values for AdaBoost +0.066, KNN +0.038, Stacking +0.029).

Elasticity: average importance on the test sample = +0.0291. Positive for 5 models: stacking (+0.145), random forest (+0.123), ridge (+0.079), AdaBoost (+0.046) and voting (+0.025). Negative only for bagging (−0.111), but this is offset by high values in other models.

Destruction stresses: average importance on the test sample = +0.01848. Most important for *k*-NN (+0.136) and bagging

(+0.113). Moderately important for random forest (+0.019). Negative for ridge (−0.018) and voting (−0.071), indicating instability.

7.4 | Reduction of Feature Space and Selection of Optimal Models

Based on the initial permutation importance analysis performed on the full set of features, it was found that the filler and residual_s-stresses features exhibit systematic noise behaviour: their permutation importance on the test sample is negative or unstable in most models, indicating that mixing these features improves or does not worsen the quality of the forecast. Thus, these features do not carry useful predictive information when extrapolating to new filler concentrations and were excluded from further analysis.

The remaining features—destruction_stresses, elasticity, viscosity, solidity and material_type—were retained in the final set

TABLE 10 | Optimized hyperparameters.

Model	Hyper params
Bagging	'bootstrap': False, 'max_features': 2, 'max_samples': 0.5, 'n_estimators': 20
decision_trees	'max_depth': 2, 'max_features': 1, 'min_samples_leaf': 1, 'min_samples_split': 2
random_forest	'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 10
gradient_boost	'learning_rate': 0.05, 'max_depth': 5, 'min_samples_leaf': 2, 'min_samples_split': 5, 'n_estimators': 10
Stacking	'dt_max_depth': 10, 'dt_min_samples_split': 2, 'final_estimator_alpha': 1.0, 'gb_learning_rate': 0.01, 'gb_max_depth': 3, 'passthrough': False, 'rf_max_depth': 5, 'rf_n_estimators': 20
ada_boost	'learning_rate': 0.1, 'loss': 'exponential', 'n_estimators': 100
Voting	'weights': [1, 2, 2]
SVM	'C': 0.1, 'epsilon': 1.0, 'kernel': 'rbf'
k-NN	'metric': 'manhattan', 'n_neighbors': 15, 'weights': 'distance'
Ridge	'alpha': 100.0, 'fit_intercept': True, 'solver': 'svd'

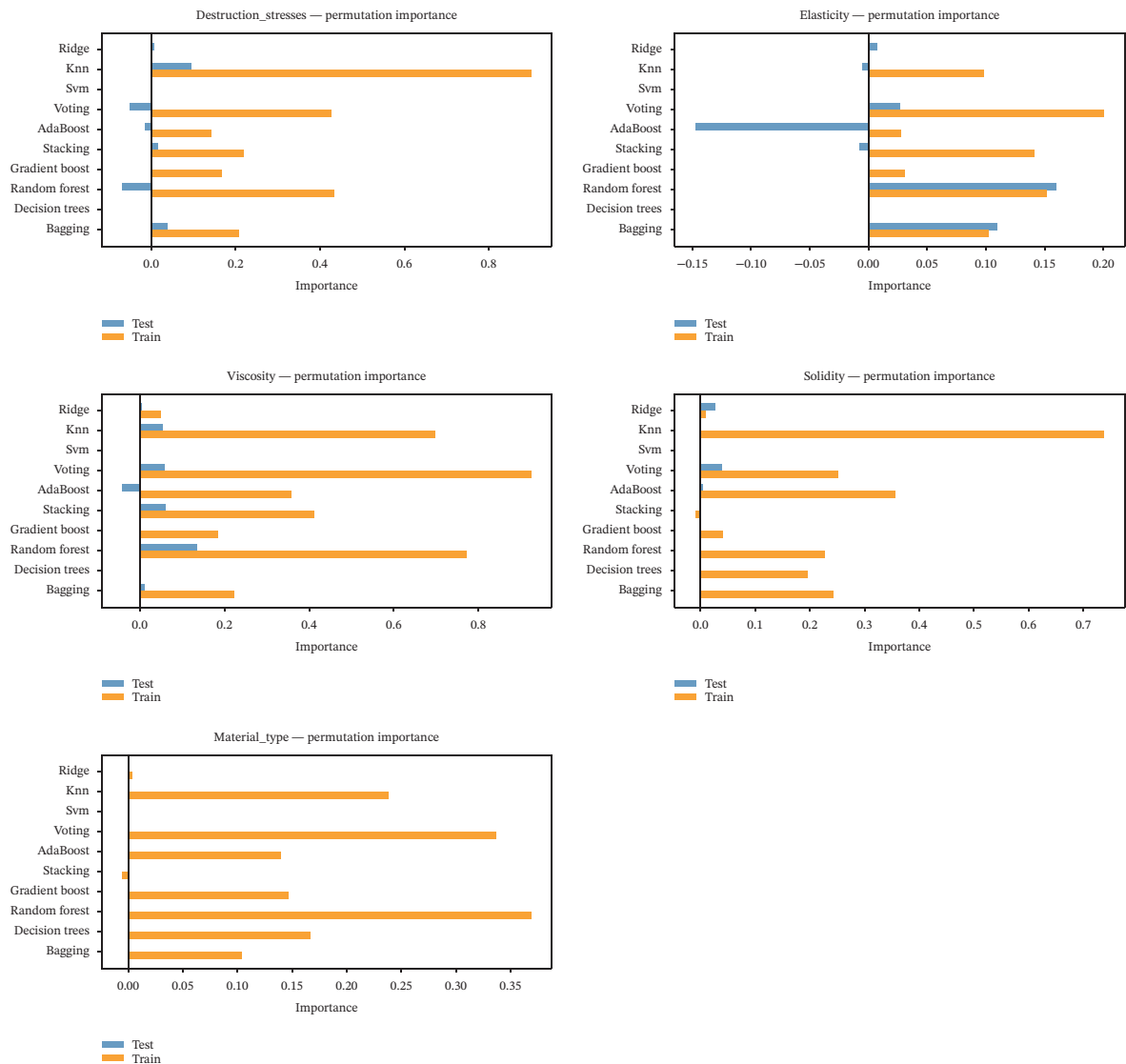


FIGURE 4 | Features permutation importance among models on train/test samples.

TABLE 11 | Comparison of MSE metric values on different samples during cross-validation and training on optimal parameters on a reduced dataset.

Model	CV-Train	CV-Val	Final-Train	Final-Val
bagging	5.511	11.3912	4.7368	9.2407
decision_trees	19.9028	9.069	17.4379	19.0
random_forest	4.081	10.4268	3.1973	39.597
gradient_boost	15.2891	13.5175	14.2407	10.2627
stacking	8.8644	9.5143	15.7106	39.3463
ada_boost	8.4342	12.511	8.1286	23.9083
voting	2.0503	10.268	1.7949	44.1188
SVM	24.1597	12.6581	21.8972	23.0687
k-NN	0.0	13.9225	0.0	12.9078
ridge	22.4613	15.7373	20.8851	14.3806

TABLE 12 | Comparison of MAPE metric values on different samples during cross-validation and training on optimal parameters on a reduced dataset.

Model	CV-Train	CV-Val	Final-Train	Final-Val
bagging	0.4672	0.5801	0.4301	0.7301
decision_trees	0.9037	0.6373	0.8272	0.9537
random_forest	0.3401	0.6093	0.3215	1.5094
gradient_boost	0.8147	0.7128	0.7765	0.6769
stacking	0.5825	0.5832	0.7191	1.5798
ada_boost	0.6739	0.6901	0.6613	1.0732
voting	0.3046	0.5745	0.2938	1.6483
SVM	0.9576	0.6318	0.8903	1.1879
k-NN	0.0	0.6761	0.0	0.8874
ridge	0.9901	0.7935	0.9294	0.9388

TABLE 13 | Comparison of MAE metric values on different samples during cross-validation and training on optimal parameters on a reduced dataset.

Model	CV-Train	CV-Val	Final-Train	Final-Val
bagging	1.6576	2.0648	1.5306	2.595
decision_trees	3.2063	2.2765	2.9357	3.4
random_forest	1.2081	2.1682	1.1417	5.37
gradient_boost	2.8885	2.5414	2.7547	2.4103
stacking	2.0648	2.0754	2.5449	5.5737
ada_boost	2.4028	2.4571	2.3583	3.8257
voting	1.0828	2.0435	1.0447	5.8626
SVM	3.3831	2.2444	3.1475	4.1891
k-NN	0.0	2.4049	0.0	3.1338
ridge	3.5113	2.8282	3.2954	3.3139

despite their varying levels of informativeness. This decision was due to the limited size of the training sample ($N_{\text{train}} = 108$): in tasks with a small number of observations, excluding features with nonnegative, albeit weak, importance can lead to the loss of useful signals and a deterioration in model quality. Therefore, the

exclusion criterion was based solely on stable negative importance in the test sample, rather than low absolute importance.

After excluding noisy features, the hyperparameters of all models $f_j, j = \overline{0, 9}$, were reoptimised using the GridSearchCV method

TABLE 14 | Comparison of R^2 metric values on different samples during cross-validation and training on optimal parameters on a reduced dataset.

Model	CV-Train	CV-Val	Final-Train	Final-Val
bagging	0.7614	0.0945	0.7773	−0.0549
decision_trees	0.1383	0.2791	0.1803	−1.1689
random_forest	0.8233	0.1712	0.8497	−3.5202
gradient_boost	0.3381	−0.0745	0.3306	−0.1715
stacking	0.6162	0.2437	0.2615	−3.4916
ada_boost	0.6349	0.0055	0.6179	−1.7293
voting	0.9112	0.1838	0.9156	−4.0364
SVM	−0.046	−0.0062	−0.0293	−1.6334
k -NN	1.0	−0.1067	1.0	−0.4735
ridge	0.0276	−0.2509	0.0183	−0.6416

with k -fold cross-validation ($k = 3$) on a reduced dataset. The optimal values of the hyperparameters determined by the search results are shown in Table 10.

A reanalysis of permutation importance was performed separately for the training and test samples, see Figure 4.

Viscosity is the dominant predictor in the training sample: train importance reaches 0.90 for voting, 0.77 for random forest and 0.70 for k -NN. At the same time, on the test sample, importance viscosity remains positive but significantly lower (bagging ~ 0.01 , random forest ~ 0.13 and stacking ~ 0.06), indicating the existence of a real physical relationship between the viscosity of the material and its heat resistance, which is partially preserved when extrapolated.

Elasticity shows the most pronounced asymmetry between train and test: for AdaBoost test, importance is -0.15 , while for bagging and random forest, it is positive (0.11 and 0.16, respectively). Note that voting shows train importance ~ 0.20 with near-zero test importance. In general, elasticity is the second most useful feature, but its impact is unstable and model dependent.

Destruction stresses has a noticeable importance on the training sample for k -NN (~ 0.90) and voting (~ 0.43), but on the test sample, its values are zero or close to zero for all models except k -NN (~ 0.13) and bagging (~ 0.04). The negative test importance for random forest (-0.07) indicates that the model misuses this feature when extrapolating.

Solidity shows high importance on the training sample for k -NN (~ 0.74), AdaBoost (~ 0.36) and voting (~ 0.25), but its test importance is zero or negative for all models without exception. This indicates that the relationship between solidity and heat resistance, learnt by the models on the training data, does not carry over to the test sample.

Material type retains nonzero importance on the training sample for most models (voting ~ 0.35 , random forest ~ 0.37 and k -NN ~ 0.24), but its test importance is zero in all cases. This is an expected result since all test samples belong exclusively to material type 4, and the feature loses its discriminatory power under extrapolation conditions.

After retraining the models on a dataset without noisy features, a comprehensive quality assessment of all ten models was performed. The results are presented in Tables 11, 12, 13 and 14.

Analysis of the MAPE metric as the main comparison criterion in extrapolation conditions allows establishing a clear hierarchy of models in terms of forecasting quality on the final test sample, as shown in Figure 5.

Gradient boost demonstrates the best result on the final test sample (Final-Val MAPE = 0.677%), which is particularly significant given that it was not the leader in the cross-validation process (CV-Val MAPE = 0.713%). It is important to note that gradient boost exhibits an atypical effect: Final-Val MAPE is lower than CV-Val MAPE, indicating that overfitting on the full Final-Train sample did not impair generalisation ability but, on the contrary, allowed the model to better learn the data structure.

The bagging model ranks second (Final-Val MAPE = 0.730%) and demonstrates the most consistent behaviour among all models: the MAPE value increases from CV-Train (0.467%) to CV-Val (0.580%) and Final-Val (0.730%), which is a sign of controlled and predictable overfitting. The gap between CV-Val and Final-Val is moderate and does not indicate a degradation in quality when retraining on the full set.

The k -NN model ranks third in terms of Final-Val MAPE (0.887%), but its behaviour is structurally anomalous: CV-Train and Final-Train MAPE are zero, which is direct evidence of complete memorisation of the training data. The competitive result on the test sample is explained not by generalisation ability but by the specificity of the distance metric at $k = 15$ neighbours with the “distance” weighting function, which actually implements weighted interpolation.

The ridge model (Final-Val MAPE = 0.939%) is the only linear model in the set and demonstrates the most stable behaviour without signs of overfitting: the MAPE values on all four samples are close (0.990/0.794/0.929/0.939%).

The most pronounced degradation in quality when moving from cross-validation to final testing is demonstrated by the random forest (CV-Val MAPE = 0.609% $\rightarrow$ Final-Val MAPE = 1.509%), voting (CV-Val = 0.575% $\rightarrow$ Final-Val = 1.648%) and stacking (CV-Val = 0.583% $\rightarrow$ Final-Val = 1.580%) models demonstrate the most pronounced degradation in quality during the transition from cross-validation to final testing. For all three models, CV-Val MAPE is one of the lowest in the set, which formally proves their effectiveness, but a significant deterioration in the final testing indicates that cross-validation overestimated their

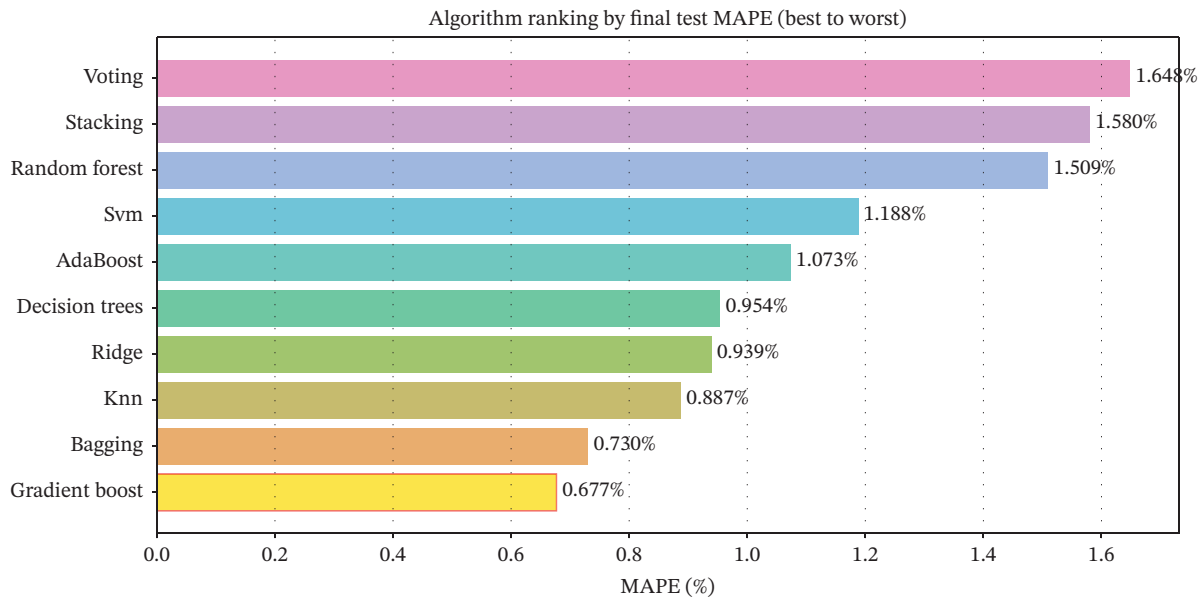


FIGURE 5 | Algorithm ranking by MAPE on Final_Test sample.

generalisation ability. This is a consequence of the fact that these models have excessive capacity and are able to adapt to a specific distribution of CV subsamples without generalising to a qualitatively different test sample.

The negative values of Final-Val R^2 for all models confirm the uninformative nature of this metric in the context of extrapolation: even the best model (gradient boost, Final-Val $R^2 = -0.172$) is formally inferior to the trivial mean model, which is an artefact of the bias of the test sample distribution relative to the training sample, rather than evidence of poor prediction quality.

7.5 | Analysis of Learning Curves on a Reduced Feature Set

The learning curves constructed after feature reduction are shown in Figure 6.

Compared with the results for the full set of features, there are a number of significant changes. After reduction, the bagging model demonstrates qualitatively different behaviour: MAE on the training and validation samples converge with an increase in sample size and stabilise at ~ 8 and ~ 20 , respectively, indicating a decrease in overfitting compared to the full set of features. Gradient boost also shows improvement: both curves gradually converge to ~ 10 – 13 MAE, although the wide confidence interval of the validation curve remains. Decision trees fundamentally change their behaviour after reduction: MAE on the training and validation samples converge and show a parallel trend at ~ 15 – 25 units, indicating a decrease in model capacity after excluding noise features.

SVM demonstrates the most stable picture among all models: both curves converge and stabilise at ~ 22 MAE, confirming the absence of overfitting. After reduction, ridge retains the characteristic convergence of curves, but the stabilisation level (~ 20 – 25 MAE) is slightly higher than for bagging and gradient boost. k -NN does not change its behaviour: MAE on the training sample remains zero throughout the range, while MAE on the

validation sample stabilises at ~ 19 , confirming structural overfitting that is not eliminated by feature reduction.

The AdaBoost and voting models remain the most prone to overfitting: the gap between MAE on the training and validation samples persists across the entire range of sample sizes, and MAE on the validation sample does not show a consistent downward trend. Stacking demonstrates the instability of both curves with wide confidence intervals, indicating high variability in estimates with small amounts of data.

7.6 | Determining the Significance of Factors Based on Statistical Tests

The paper also assesses the importance of the influence of the values of the variables: filler, destruction stresses, elastic modulus, impact strength, adhesive strength (solidity), residual stresses and material type on the value of the variable heat resistance based on statistical tests. In particular, Pearson's correlation statistical test and Fisher's one-way F-test (one-way ANOVA) were used. Considering y from Formula (5) as the variable heat resistance and $x_0, x_1, x_2, x_3, x_4, x_5$ and x_6 as variables filler, destruction stresses, elastic modulus, impact strength, adhesive strength (solidity), residual stresses and material type, respectively, these statistical tests analyse exclusively one-dimensional linear dependencies between a separate variable x_i and variable y .

The statistical test for Pearson's correlation was calculated using the following formula:

$$r_i = \frac{\text{cov}(x_i, y)}{\sigma_{x_i} \cdot \sigma_y}, \quad (6)$$

where: r_i is Pearson's correlation coefficient; $\text{cov}(x_i, y) = (1/(M-1)) \sum_{m=0}^{M-1} (x_{im} - \bar{x}_i)(y_m - \bar{y})$ is the covariance, x_{im} and y_m are the values of the variables in the sample, $\bar{x}_i$ and $\bar{y}$ are their mean values, σ_{x_i} is the standard deviation of the variable x_i

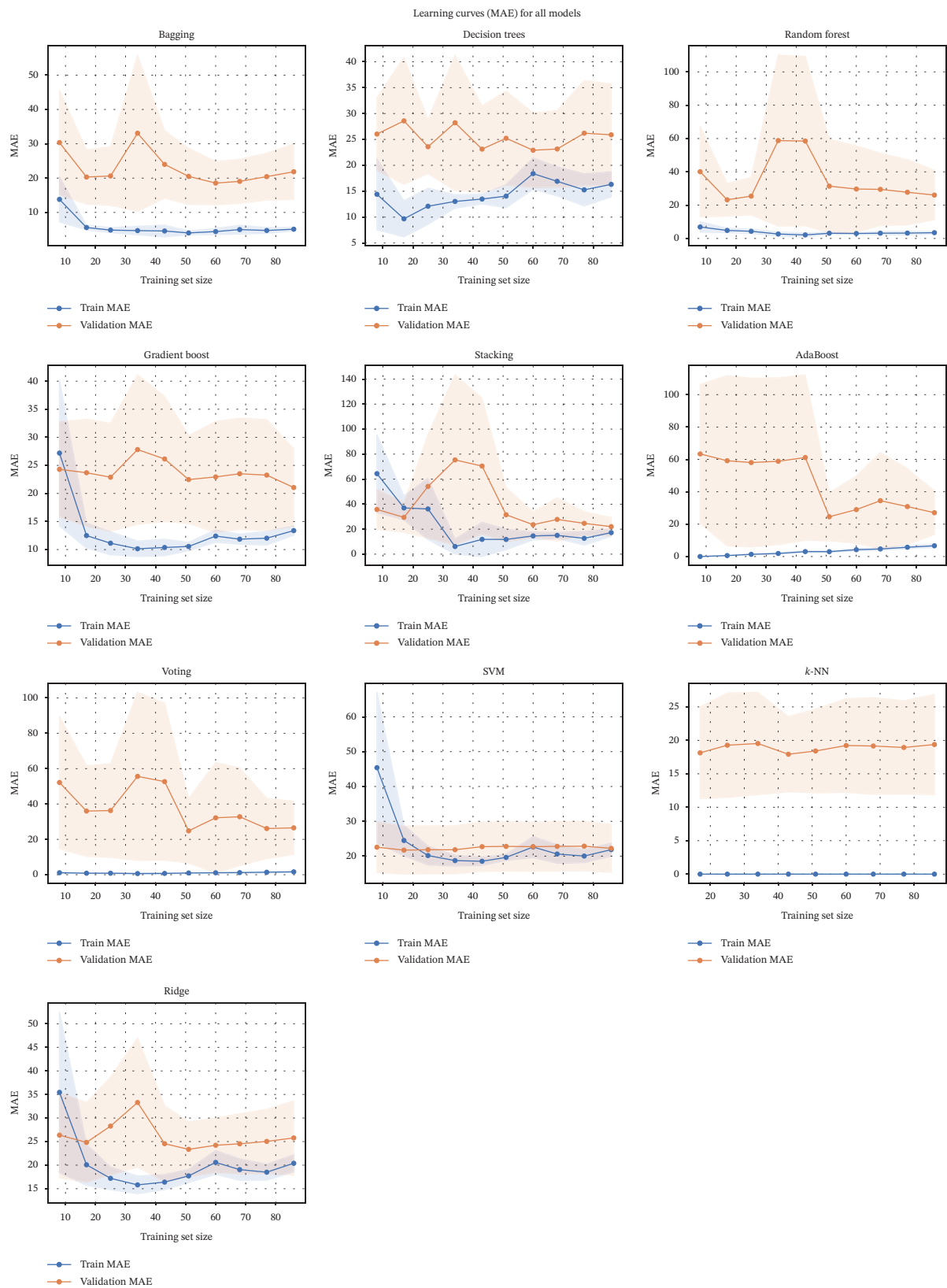


FIGURE 6 | Learning curves on a reduced feature dataset.

TABLE 15 | Results of the statistical test for Pearson correlation.

Feature	Correlation coefficient	<i>p</i> value ($\alpha = 0.05$)
Filler	0.1	0.27
Destruction stresses	0.057	0.55
Elastic modulus (elasticity)	−0.02	0.76
Impact strength (viscosity)	0.114	0.23
Adhesive strength (solidity)	−0.071	0.46
Residual stresses	−0.09	0.35
Material type	−0.108	0.26

TABLE 16 | Results of the one-factor Fisher's *F*-test.

Feature	F-score	<i>p</i> value ($\alpha = 0.05$)
Filler	1.22	0.27
Destruction stresses	0.34	0.55
Elastic modulus (elasticity)	0.09	0.76
Impact strength (viscosity)	1.4	0.23
Adhesive strength (solidity)	0.53	0.46
Residual stresses	0.86	0.35
Material type	1.26	0.26

TABLE 17 | Mutual information.

Feature	Info value	Conclusion on importance
Destruction stresses	0.39	Critical variable (highest information value)
Impact strength (viscosity)	0.13	Medium importance (significant information value)
Filler	0.09	Moderate importance (lower but useful information value)
Adhesive strength (solidity)	0.06	Low importance (limited information value)
Residual stresses	0.01	Very low importance (minimal information value)
Material type	0.006	Slightly/almost uninformative (may be considered for deletion)
Elastic modulus (elasticity)	0	Uninformative (completely independent of Heat resistance variable. Recommended for deletion)

and σ_y is the standard deviation of the variable *Y*, using the “r_regression” function from scikit-learn v1. 7.2 library [49].

The results are shown in Table 15.

From the results of the Pearson correlation test, it can be concluded that there is no significant linear dependence of the values of the heat resistance variable on the values of the variables: filler, destruction stresses, elastic modulus, impact strength, adhesive strength, residual stresses and material type, since the *p* value is > 0.05 —hypothesis H_0 is met.

Let us move on to the Fisher's one-way ANOVA.

The one-way ANOVA was calculated using the formula in (3):

$$F_i = \frac{r_i^2}{1 - r_i^2} \cdot (M - 2), \quad (7)$$

where *M* is the number of observations and r_i is the Pearson correlation coefficient between variables x_i and *y*, using the “f_regression” function from the scikit-learn v1.7.2 library [50, 51].

The results are shown in Table 16.

From the results of the Fisher's one-factor *F*-test, it can be concluded that there is no significant linear dependence of the values of the heat resistance variable on the values of the variables: filler, destruction stresses, elastic modulus, impact strength, adhesive strength, residual stresses and material type are not significant since the *p* value > 0.05 —hypothesis H_0 is fulfilled.

8 | Determining the Importance of Factors Based on Information Measures

In order to assess the importance of the influence of the values of the variables: filler, destruction stresses, elastic modulus, impact strength, adhesive strength, residual stresses and material type on the value of the variable heat resistance, we also used an information measure based on Shannon's entropy, namely, Mutual Information. The numerical values for the Mutual Information measure were calculated using the following formula:

$$I(x_i; y) = H(x_i) + H(y) - H(x_i, y), \quad i = \overline{0, 6} \quad (8)$$

where $I(x_i; y)$ is the mutual information between x_i and y ; $H(x_i)$ is the Shannon entropy of the variable x_i ; $H(y)$ is the Shannon entropy of the variable y and $H(x_i, y)$ is the joint Shannon entropy.

Numerical values for Mutual Information were calculated using the “mutual_info_regression” function from the scikit-learn v1.7.2 library [52]. Larger numerical values of Mutual Information for a variable x_i indicate that this variable has a greater influence on y . Mutual Information can detect nonlinear relationships between x_i and y . The results are shown in Table 17.

Analysing Mutual Information, we can draw the following conclusions:

1. The variable destruction stresses has the greatest impact on the variable heat resistance. Thus, by changing the value of destruction stresses, we can significantly change the value of the variable heat resistance. In further research of new models (ML algorithms), we can use it primarily to predict the value of the variable heat resistance;
2. Variables with an “info value” (Table 17) that are less than 0.01, namely, material type and elastic modulus, can be excluded in further research on new models (ML algorithms) to simplify them without significantly losing the accuracy of predicting the values of the heat resistance variable.

It is important to note that Mutual Information evaluates the importance of each variable $x_0, x_1, x_2, x_3, x_4, x_5$ and x_6 in isolation, without taking into account their interaction. That is, for example, the synergy of two variables that together explain y better than each separately is not taken into account.

9 | Conclusions

A comprehensive analysis of permutation importance, training curves on a reduced feature set and mutual information estimates allows us to draw conclusions about the importance of features and the selection of optimal models for predicting the target feature.

According to the results of the Mutual Information analysis, the feature destruction_stresses has the greatest isolated impact on the target variable heat_resistance, making it a priority predictor when building simplified models in further research. At the same time, the features material_type and elasticity demonstrate an info value below the threshold level of 0.01, which indicates their low individual informativeness and allows them to be excluded from the feature set without a significant loss of prediction accuracy. In particular, the permutation importance results indicate that elasticity, despite its low individual info value, retains positive importance on the test sample for bagging and random forest models, which may be a result of interaction with other predictors. Thus, decisions to exclude features with low info value should be made taking into account the results of permutation importance and not solely on the basis of Mutual Information.

Taking into account both methods of assessing the importance of features, the most reliable predictors for the task of predicting heat resistance are destruction_stresses—as the dominant

predictor according to Mutual Information—and viscosity, which demonstrates the highest permutation importance on the training sample and maintains a consistently positive importance when extrapolated. The features solidity and material_type, despite their usefulness within the training sample, do not demonstrate consistent generalization ability when extrapolated to new material compositions.

Based on the combined results of all analyses, the models considered are divided into three groups. The group of optimal models includes gradient boost (Final-Val MAPE = 0.677%) and bagging (Final-Val MAPE = 0.730%), which demonstrate the best absolute accuracy on the test sample, an acceptable level of overfitting on the training curves and predictable behaviour when transitioning from cross-validation to final testing. The group of acceptable models includes ridge as the most resistant to overfitting model with a minimum gap between CV-Val and Final-Val MAPE (0.794% and 0.939%, respectively), as well as decision trees, whose behaviour has significantly improved after feature reduction. The group of models with unsatisfactory generalization ability includes random forest, voting, stacking, AdaBoost, SVM and k -NN, which either demonstrate persistent overfitting and a sharp degradation in quality during final testing or do not provide sufficient accuracy when extrapolating to new material compositions.

The information obtained is potentially valuable for experimenters in terms of optimally controlling the importance values of variables and organizing experiments to create new materials with predictable values.

Funding

No funding was received for this manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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